Band-1 Hadamard order 14740 – Classical Paley-II DIRECT (q2=7369); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14740 Paley-II PASS; Paley-I FAIL/SKIP (q=14739=3*17^3 NOT prime); Kronecker FAIL/SKIP (4 3685; 3685=5*11*67 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-14

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-14_Band1_Hadamard_14732_PaleyI` (current bookkeeping); `2026-10-14_Band1_Hadamard_147` `2026-10-14_Band1_Hadamard_14700_PaleyII` (Paley-II construction template); `2026-10-13_Band1_Hadamard_14` (shape peer ===4/m===1); **Lab.** `/workspace/experiments/band1/hadamard_14740_paleyII/`.
Cutoff. 2026-10-14T16:42:44Z (~12:42PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14740 only (verified max|HH^T-nI|=0), via classical Paley-II q2=7369 (prime ===1 mod 4), order 2(q2+1). 14740 IN SPK via Paley-II. 14740 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14740: Paley-II **PASS** ($q_2 = 7369$ prime $\equiv 1 \pmod 4$, $n = 2(q_2 + 1)$); Paley-I **FAIL/SKIP** ($q = 14739 = 3 \cdot 17^3$ NOT prime)

Abstract

Finite construction of order **14740** (eleventh Paley-II DIRECT wave of the 14xxx chain, after 14356, 14428, 14460, 14476, 14508, 14596, 14620, 14644, 14668 and 14700).

- H_{14740} = Paley-II ($q_2 = 7369$) – classical SPK
- Paley-I for 14740: **FAIL/SKIP** ($q = 14739 = 3 \cdot 17^3$ NOT prime)
- Kronecker for 14740: **FAIL/SKIP** ($4 \cdot 3685$; $3685 = 5 \cdot 11 \cdot 67$ odd composite, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14740.py`

Results. overall_pass=True; units 932/932; row blocks 116/116 (block size 128); wall 6335.4321s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 14740 = 2(q_2 + 1)$ with $q_2 = 7369$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14739 = 3 \cdot 17^3$ NOT prime — FAIL/SKIP.
- Kronecker: $14740 = 4 \cdot 3685$; $3685 = 5 \cdot 11 \cdot 67$ odd composite ($\equiv 1 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14740 = 2^2 \cdot 5 \cdot 11 \cdot 67 \equiv 4 \pmod{16}$, $m = 3685 \equiv 1 \pmod 4$ odd NOT-admissible (SHAPE-CLASS CHANGE from template $14700 \equiv 12/m \equiv 3$; shape peer $14644 = 2^2 \cdot 7 \cdot 523 \equiv 4 \pmod{16}$, $m \equiv 1$).
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7369}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7370$ gives

$$H_{14740} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14740 Paley-II	Paley-II q2=7369	0	6316.1032
H_14740 Paley-I	FAIL/SKIP (q=14739 3*17ˆ3 NOT prime)	–	–
H_14740 Kronecker	FAIL/SKIP (4*3685; 3685=5*11*67 odd NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14740 in SPK	**True** (Paley-II q2=7369)
Prior Paley-I 14732	IN SPK via Paley-I q=14731 (~10:51AM ET 2026-10-14)
Prior Paley-I 14724	IN SPK via Paley-I q=14723 (~9:03AM ET 2026-10-14)
Prior Paley-II 14700	IN SPK via Paley-II q2=7349 (~6:50AM ET 2026-10-14)
Prior Paley-I 14684	IN SPK via Paley-I q=14683 (~1:20AM ET 2026-10-14)
Prior Paley-II 14668	IN SPK via Paley-II q2=7333 (~10:29PM ET 2026-10-13)
Prior Paley-II 14644	IN SPK via Paley-II q2=7321 (~8:44PM ET 2026-10-13)
Prior Paley-I 14640	IN SPK via Paley-I q=14639 (~6:45PM ET 2026-10-13)
Prior Paley-I 14628	IN SPK via Paley-I q=14627 (~4:42PM ET 2026-10-13)
Prior Paley-II 14620	IN SPK via Paley-II q2=7309 (~3:13PM ET 2026-10-13)
Prior Paley-II 14596	IN SPK via Paley-II q2=7297 (~1:12PM ET 2026-10-13)

Quantity	Value
Prior 14592	IN SPK via Paley-I q=14591 (~11:06AM ET 2026-10-13)
Prior 14432	IN SPK via Paley-I (~7:51PM ET 2026-10-09)
Prior Paley-II 14428	IN SPK via Paley-II q2=7213 (~7:10PM ET 2026-10-09)
Prior Paley-II 14356	IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14740 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14740 (FIRST any direct = Paley-I): q=14747->n=14748 Paley-I DIRECT (prime ===3 mod 4; Paley-II q2=7373=73*101 FAIL; Kronecker 14748=4*3687; 3687=3*1229 odd (===3 mod 4) NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14732_paleyI / hadamard_14724_paleyI) | Kronecker-only NOTE 14688=4*3672 (H_3672 via Paley-I q=3671; no direct Paley route; note-only, unbuilt) | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; below 14740, note-only, unbuilt) | skipped Paley-II DIRECT orders below 14740 still unbuilt (template hadamard_14740_paleyII / hadamard_14700_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)